Add, Don't Remove: Where Target Supervision Helps in Text–Tabular Learning

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Abstract

In the embedding baselines of the MulTaBench text–tabular benchmark, each free-text column is embedded, compressed with PCA, and given to a tree model together with the table; Joint TAR first fine-tunes the text encoder on the target, making the embeddings target-aware before compression. We ask how the target can be used further on top of TAR. Our first approach *removes* information: a sparse autoencoder (SAE) splits each column's embeddings into concepts, each concept is tested against a shuffled copy of itself inside one model with the table and all other concepts, and the concepts that fail are subtracted before compression. On the 20 text datasets of MulTaBench this selection is far better than removing random concepts of the same total variance (16/20 datasets, $p < 0.001$), yet it does not improve on TAR, not even when the amount removed is chosen by nested cross-fitting. Approaches that *reweight* the embedding or re-learn its compression (PLS instead of PCA, no compression, a 30-unit bottleneck trained on the target) do not help either. What helps is *adding*: the out-of-fold predictions of a text-only model, stacked as extra columns next to TAR's, improve on TAR with tuned LightGBM ($+0.0035$, 16/20, Holm-adjusted $p = 0.007$) and with CatBoost ($+0.0036$, 17/20, $p = 0.004$). With default LightGBM it wins on 15/20 datasets but is not significant, because the default learner fails on two folds of one 114-class dataset where the other learners, given the same features, do not.

1 Introduction

Many tabular datasets contain free-text columns next to numerical and categorical ones, such as the description of a job posting or the text of a product review. A standard pipeline embeds each text cell with a pretrained encoder, reduces it to a few dimensions, and gives the result to a gradient-boosted tree model alongside the other columns [7, 10]. Among its baselines, MulTaBench [12] includes two such pipelines: *Joint Frozen* embeds each text column with a frozen E5-small encoder [13] and keeps 30 principal components per column; *Joint TAR* first fine-tunes the encoder on the target, with the aim of making the embeddings, and therefore the components, target-aware.

Both pipelines must compress each column to 30 dimensions, and PCA decides what survives by variance, not by relevance. Fine-tuning shifts the embedding toward the target, but it does not control the compression: if a frequent but irrelevant theme—shipping, brand tone—varies more than the price-relevant content of a product description, it still takes up the components. Our goal is therefore to remove target-irrelevant content *before* compressing, so that what PCA discards is the least relevant part. Sparse autoencoders (SAEs) make this possible by writing an embedding as a sparse combination of learned, often interpretable directions [3, 5, 6]. We use them as a filter, and compare removal with two other ways of using the target on top of TAR: reweighting the embedding, and adding a text model's prediction. Our contributions are:

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- SAE concept pruning for text–tabular pipelines, with a joint shadow test of every concept and a nested cross-fitted variant that makes every decision on rows the encoder never saw (§3).
- A paired, leakage-controlled evaluation on all 20 MulTaBench text datasets with three learners and Holm-corrected tests, in which every arm of a fold uses the same saved TAR fine-tune (§4).
- The finding that removal and reweighting do not improve on TAR while adding does: out-of-fold text stacking on TAR is significantly better than TAR with tuned LightGBM and with CatBoost (§5).

2 Related work

Text in tabular learning. Benchmarks of tables with text fields [10, 12] show that tree ensembles over compressed text embeddings are strong baselines. Target-aware text representations appear in fine-tuning the encoder before compression (Joint TAR, 12) and in tabular foundation models that attend from target tokens to verbalised cells, such as TabSTAR [1].

Using the target before or beside the learner. Supervised PCA screens features by their correlation with the outcome before extracting components [2]; PLS finds directions of maximal covariance with the target [14]. Stacked generalization feeds out-of-fold base-model predictions to a second learner [15]; cross-fitting [4] is the same principle for nuisance estimates.

Feature relevance and SAEs. Marginal importance is unreliable for redundant features [11]; Boruta [9] compares each feature with permuted "shadow" copies inside one model. SAEs are a central tool of mechanistic interpretability [3, 5]; we use TopK SAEs [6] to filter representations for a downstream predictor.

3 Method

Pipeline. For each text column c, $x \in \mathbb{R}^{384}$ is the (frozen or TAR-tuned) E5 embedding of "`passage: c: value`". Joint Frozen and Joint TAR fit PCA-30 on the training rows and pass the 30 components to the tree model together with the table. Every method below is fitted on the training rows only.

Removing: concept pruning. A TopK SAE [6] with 2048 latents and $k = 32$ is trained, without labels, on the column's training embeddings: $\hat{x} = b + \sum_i z_i d_i$ with unit-norm directions d_i. Concepts active on fewer than 20 rows are always kept. The others are clustered by decoder direction into sibling groups, and one LightGBM is fitted on [table, all concepts, a jointly row-permuted "shadow" copy of every group], three times; a group is dropped unless its split gain beats its own shadow's in at least two repeats. Dropped concepts are subtracted, $\tilde{x} = x - \sum_{i \in \mathcal{D}} z_i d_i$, and PCA-30 is fitted on $\tilde{x}$. In the *cross-fitted* variant the amount is chosen instead: the training rows are split into three parts; for each, TAR is fine-tuned on the other two, concepts are ranked there by variance per unit of shadow-corrected gain, and the removal fractions $\{0, 10, 25, 50, 75\}\%$ are scored on the unseen part with a LightGBM on the final inputs. The best fraction (0% included) is applied to the fold's TAR.

Reweighting and learned compression. PLS-30 [14] on TAR embeddings in place of PCA-30; no compression (all 384 dimensions); and a *learned* compression: TAR's fine-tuning recipe with a 30-unit linear bottleneck between the pooled embedding and the head (bottleneck, GELU, linear head), whose 30 outputs per column replace PCA-30, so that the compression itself is trained on the target.

Adding: text stacking. An MLP (one hidden layer of 256, dropout 0.3, early stopping) is trained on the concatenated frozen E5 embeddings of all text columns to predict the target. Its predictions (one logit, one log-probability per class, or one value) are added as columns next to TAR's PCA columns. To avoid leakage they are computed out of fold: five folds over the training rows, rows with identical text in the same fold, each training row scored by the model that did not see it; validation

Table 1: Arms that change what follows TAR's encoder, per learner: mean difference from that learner's Joint TAR, datasets improved (of 20), Wilcoxon p. Default-LightGBM Joint Frozen is in the main run (-0.017, 1/20).

Arm (Δ vs. TAR, datasets won, p)	LightGBM	LightGBM tuned	CatBoost
Joint Frozen	–	-0.0140, 1/20, <0.001	-0.0150, 1/20, <0.001
TAR + text stacking	-0.0020, 15/20, 0.064	+0.0035, 16/20, 0.001	+0.0036, 17/20, <0.001
TAR + PLS-30	-0.0038, 6/20, 0.040	-0.0057, 3/20, <0.001	-0.0065, 3/20, <0.001
TAR, learned bottleneck	-0.0058, 6/20, 0.015	-0.0079, 3/20, <0.001	-0.0061, 4/20, <0.001

Table 2: Final run, default LightGBM: every arm vs. Joint TAR with the same fine-tuned encoder (per-dataset means over 6 folds); p: Wilcoxon over datasets; p_{Holm}: adjusted within this table.

Method	Δ vs. TAR	datasets won	folds won/lost	p	p_{Holm}
TAR + text stacking	-0.0020	15/20	85/35	0.064	0.191
TAR + PLS-30 (instead of PCA)	-0.0038	6/20	44/76	0.040	0.160
TAR, learned 30-d bottleneck	-0.0058	6/20	39/81	0.015	0.092
TAR, all 384 dims	+0.0014	9/20	63/57	0.729	0.729
Late fusion (frozen text model)	-0.0181	6/20	39/81	0.030	0.148
TAR + random pruning (variance-matched)	-0.0106	2/20	30/90	<0.001	<0.001
TAR + concept pruning (ours)	-0.0027	8/20	48/72	0.216	0.432
TAR + random pruning	-0.0217	1/20	19/101	<0.001	<0.001

and test rows get the average of the five models. *Late fusion* uses the same text model differently: a logistic/linear meta-learner combines its out-of-fold outputs with those of a table-only LightGBM.

4 Experimental setup

Data and protocol. All 20 text datasets of MulTaBench (3 binary, 7 multiclass, 10 regression), with the benchmark's splits (two repeats of 3-fold cross-validation; training sets capped at 10k rows) and metrics: AUROC (one-vs-rest macro for multiclass) and R^2. Joint TAR follows the benchmark recipe: LoRA [8] of rank 16 on the last three E5 layers, a linear head, regression targets binned into 20 quantiles, early stopping on 10% of the training rows. In the final run each fold's LoRA adapters are saved and restored for every arm and learner, so all comparisons within a fold share one fine-tune.

Learners. LightGBM with default settings (as in the benchmark); LightGBM tuned on its validation split (learning rate, depth, column subsampling; 12 configurations with early stopping); and CatBoost with the benchmark's defaults. Every arm, including TAR, is evaluated with each learner.

Controls and statistics. Random pruning removes random eligible concepts from the same SAE, either the same *number* as concept pruning or the same *variance*. Each arm is compared with Joint TAR of the same learner and fold by a two-sided Wilcoxon signed-rank test over per-dataset means; p-values are Holm-adjusted over all fourteen arm–learner comparisons with TAR.

5 Results

Adding a text model's prediction improves on TAR. With tuned LightGBM, TAR + stacking beats TAR by $+0.0035$ on 16/20 datasets (Holm-adjusted $p = 0.007$); with CatBoost by $+0.0036$ on 17/20 ($p = 0.004$; Table 1). Both learners are themselves stronger than default LightGBM (their TAR is $+0.012$ and $+0.014$ higher), so the gain is not an artefact of a weak learner. With default LightGBM, stacking improves 15/20 datasets but the mean is negative (-0.0020, $p = 0.064$): on two folds of Spotify genres (114 classes) the default learner collapses to chance level (test log loss 33), whereas tuned LightGBM and CatBoost, given exactly the same stacked features on those folds, score 0.95–0.97, above their TAR. The failure is therefore in fitting default LightGBM to these features, not in the features; we report it unchanged.

Table 3: What is removed from TAR embeddings, averaged over datasets, folds and text columns.

Removal	concepts removed	variance removed	removed groups never used
Concept pruning (ours)	70%	38%	23%
Random, same number	70%	57%	–
Random, same variance	53%	39%	–

Reweighting does not help. PLS in place of PCA is worse than TAR with every learner (-0.004 to -0.007; significant after correction with tuned LightGBM and CatBoost), and keeping all 384 dimensions is indistinguishable from TAR ($+0.0014$, 9/20; Table 2). This last arm tests the premise of removal: if compressing TAR embeddings to 30 dimensions discarded much target-relevant information, keeping all 384 would win. It does not, so after fine-tuning there is little for a better compression to recover. Learning the compression does not help either: the 30-unit bottleneck is worse than TAR with every learner (-0.006 to -0.008, better on 3–6 datasets), almost entirely on binary and regression datasets (multiclass: ≤ 0.001). A plausible reason is that it keeps only what separates the training classes—for regression, the 20 bins—and discards finer information that PCA on TAR's embedding retains. Late fusion helps on multiclass datasets (6/7) but loses on all binary and regression datasets (-0.018 overall): averaging a table model and a text model discards the interactions a joint learner can use.

Removing does not help, although its selection is meaningful. Concept pruning on TAR is not better than TAR (-0.0027, 8/20, $p = 0.22$; in an earlier full run -0.0035, $p = 0.006$). It removes about 70% of the eligible concepts and 38% of the embedding variance (Table 3), and the selection is meaningful: random removal of the same variance is worse by 0.0078 (16/20, $p < 0.001$), and random removal of the same number of concepts, which takes out 57% of the variance, by 0.019. 23% of the dropped groups were never split on by either the concept or its shadow, so "removed" partly means "unused". Sibling grouping rarely merged concepts, so redundant concepts share gain inside the joint model rather than being tested as a unit. Making every decision on unseen rows does not change the outcome: the cross-fitted variant (five datasets, Table 6) ties TAR (-0.0003), choosing 0% where removal hurts and 75% on two datasets where removal helped the inner encoders (trained on two thirds of the rows) but not the final one.

6 Discussion, limitations and conclusion

After target-aware fine-tuning, TAR's 30 components already concentrate on target-relevant directions: deleting the rest costs little but frees nothing, and re-projecting them with PLS does not help. Our interpretation is that a separately trained text model's prediction is information in a form the tree model cannot easily build from 30 components per column, so it adds to TAR as long as the learner is robust. The stacked text model is deliberately simple (frozen E5, MLP); the gain is small (≈ 0.0035) but appears in every task type with both robust learners (binary 3/3 datasets with both; multiclass 6/7 and 5/7; regression 7/10 and 9/10).

Limitations. One encoder (E5-small) and one SAE size. A pilot with TabSTAR [1] as backbone was infeasible on our hardware and is not reported. Late fusion trains on all training rows while the other arms set aside the learner's validation split. Cross-fitting was run on five datasets only. With default LightGBM, five multiclass runs collapsed to chance level (stacking on Spotify, two folds in each of two runs; one held-out pruning run on Wine); the evidence above points to the learner, but we did not identify the exact mechanism.

Code. `https://github.com/fireflyOr/NLP-final-project`. All results can be regenerated with `sae_pilot.py`; tables with `paper/make_tables.py`.

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A Per-dataset results

Table 4: Mean AUROC (classification) or R^2 (regression) over 6 folds, LightGBM. Best per row in bold. †: fewer than 6 folds.

Dataset	Frozen	TAR	TAR+prune	TAR+rand	PLS	Stack
Fake job postings (bin)	0.914	0.911	0.915	0.903	0.907	**0.920**
Jigsaw toxicity (bin)	0.819	**0.933**	0.929	0.880	0.864	0.864
Kickstarter (bin)	0.718	**0.734**	0.728	0.708	0.724	0.727
DS salary (mul)	0.798	**0.807**	0.803	0.797	0.793	0.806
Michelin (mul)	0.893	**0.904**	0.903	0.901	0.877	0.902
Product sentiment (mul)	0.874	**0.890**	0.887	0.889	0.846	0.879
Spotify genres (mul)	0.900	0.904	**0.905**	0.903	0.900	0.794
US accidents (mul)	0.970	**0.978**	**0.978**	0.975	0.970	0.973
Wine reviews (mul)	0.966	**0.975**	**0.975**	0.974	0.971	0.974
Women's clothing (mul)	0.898	**0.906**	0.904	0.903	0.896	0.902
Baby product prices (reg)	0.777	0.795	0.789	0.757	**0.808**	0.808
Book prices (reg)	0.496	0.514	0.499	0.469	0.501	**0.521**
Book readability (reg)	0.805	**0.815**	0.813	0.803	0.809	0.807
Mercari (reg)	0.414	0.439	0.433	0.409	0.437	**0.445**
Montgomery salaries (reg)	0.973	0.975	0.975	**0.976**	0.975	0.974
Rotten Tomatoes (reg)	0.494	0.506	0.500	0.474	0.480	**0.509**
SCImago impact (reg)	0.843	**0.849**	0.847	0.843	0.839	0.845
Vancouver salaries (reg)	0.734	0.765	0.768	0.756	**0.769**	0.766
Video game sales (reg)	0.368	0.379	0.355	0.255	0.369	**0.384**
Zomato (reg)	0.783	0.802	0.802	0.788	0.791	**0.809**

Table 5: Final run, LightGBM: mean score per dataset for each TAR-based arm.

Dataset	TAR	+stack	+PLS	384-d	BN-30	Late	+prune	+rand(var)
Fake job postings (bin)	0.909	0.909	0.908	**0.923**	0.895	0.906	0.914	0.903
Jigsaw toxicity (bin)	0.932	0.932	0.924	**0.936**	0.932	0.881	0.929	0.927
Kickstarter (bin)	**0.734**	0.733	0.729	0.732	0.729	0.726	0.730	0.714
DS salary (mul)	0.807	0.809	0.796	0.800	0.806	**0.827**	0.803	0.804
Michelin (mul)	0.903	0.906	0.894	0.903	0.903	**0.910**	0.905	0.903
Product sentiment (mul)	0.890	0.896	0.878	0.889	0.891	**0.910**	0.890	0.889
Spotify genres (mul)	0.904	0.794	0.904	0.895	0.904	**0.944**	0.906	0.905
US accidents (mul)	0.978	**0.979**	0.977	**0.980**	0.977	0.969	0.979	**0.979**
Wine reviews (mul)	0.975	0.977	0.975	0.974	0.973	**0.980**	0.974	0.974
Women's clothing (mul)	0.905	0.905	0.902	**0.907**	0.905	**0.907**	0.905	0.905
Baby product prices (reg)	0.788	**0.806**	0.799	0.802	0.799	0.766	0.793	0.770
Book prices (reg)	0.512	**0.516**	0.499	**0.516**	0.504	0.465	0.499	0.484
Book readability (reg)	0.816	**0.817**	0.812	0.810	0.807	0.802	0.812	0.813
Mercari (reg)	0.442	**0.453**	0.439	0.441	0.431	0.422	0.431	0.419
Montgomery salaries (reg)	0.975	0.975	**0.976**	0.973	**0.976**	0.966	0.975	0.975
Rotten Tomatoes (reg)	0.504	**0.516**	0.482	0.510	0.495	0.496	0.494	0.493
SCImago impact (reg)	0.849	**0.850**	0.840	0.844	0.844	0.834	0.847	0.845
Vancouver salaries (reg)	0.767	0.766	0.774	**0.781**	0.768	0.649	0.770	0.764
Video game sales (reg)	0.384	0.374	**0.386**	0.370	0.325	0.297	0.361	0.300
Zomato (reg)	0.801	0.819	0.807	**0.820**	0.796	0.757	0.804	0.798

Table 6: Nested cross-fitted removal amount (five datasets). Every decision is made on rows the judging encoder never saw.

Dataset	TAR	TAR + cross-fitted pruning	chosen removal (per fold)
Fake job postings	0.912	0.911	75, 75, 75, 75, 75, 75 %
Jigsaw toxicity	0.931	0.927	75, 75, 75, 75, 75, 75 %
Kickstarter	0.734	0.734	0, 0, 0, 0, 0, 0 %
DS salary	0.807	0.807	0, 0, 0, 0, 0, 0 %
Baby product prices	0.789	0.794	50, 50, 10, 25, 0, 25 %